Lab 3: Architectural Pattern Justification

System: Telemedicine Slot Booking & Prescription Portal

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Architecture Selection: "We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal System."

Reason 1 — Independent Scaling

Appointment booking traffic spikes sharply during peak hours (mornings, evenings, and cold/flu season), while services such as prescription generation and notifications see steadier, lower demand. A microservices approach lets the Appointment Booking Service scale out independently on additional instances during peak load, without needlessly scaling the Prescription Service, Payment Service, or Notification Service, which keeps infrastructure cost proportional to actual demand.

Reason 2 — Fault Isolation for a Critical Healthcare Workflow

In a telemedicine platform, some functions are safety-critical (booking a consultation, issuing a prescription) while others are supporting (SMS/email reminders, payment receipts). With microservices, if the Notification Service or Payment Service experiences an outage or slowdown, patients can still book slots, and doctors can still issue prescriptions, because each service runs and fails independently. A monolithic or tightly layered design would risk one failing module bringing down the entire consultation workflow.

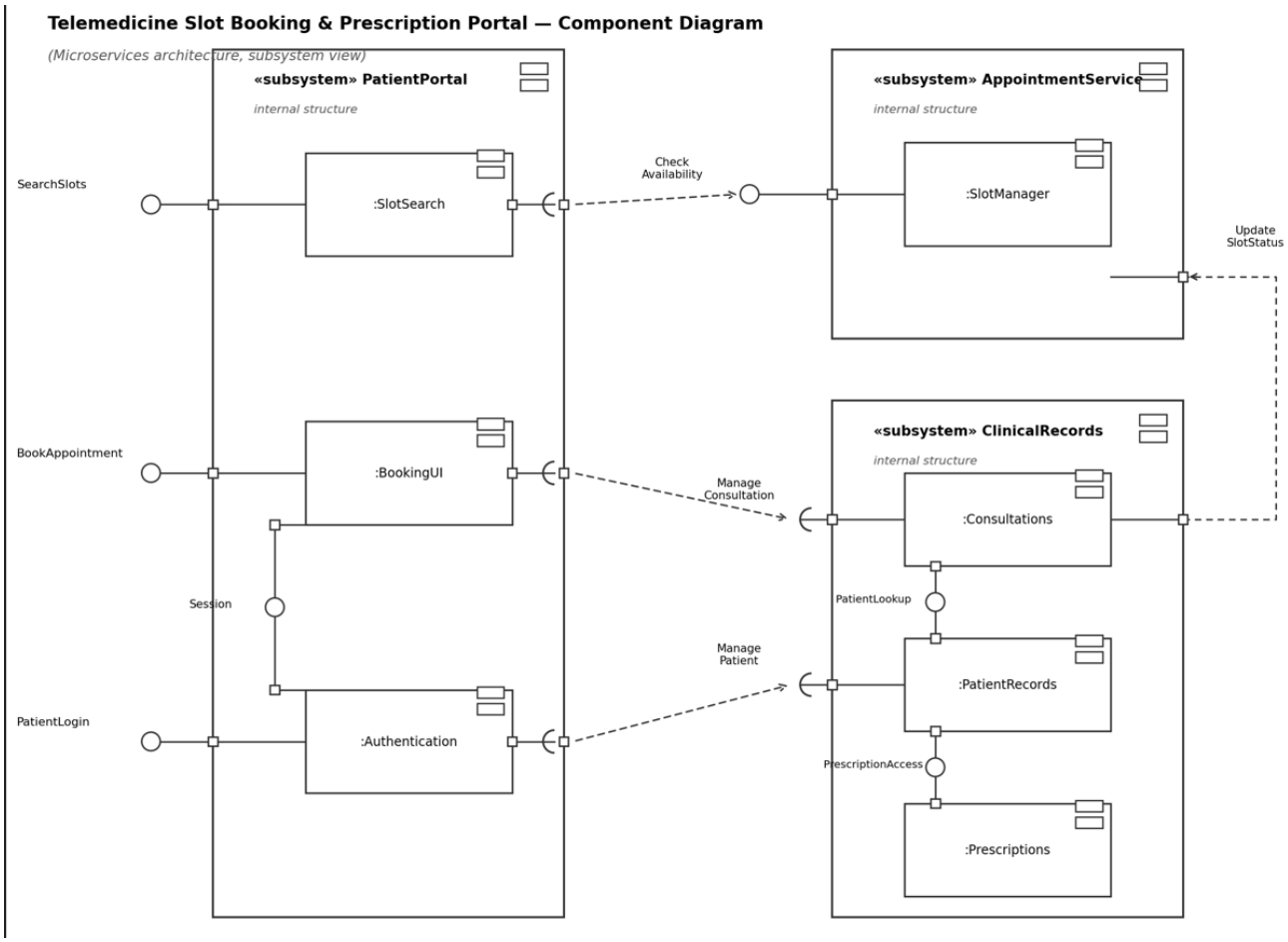
Security Advantage

Splitting the system into services allows patient health data (medical history, prescriptions) to be isolated inside the Prescription Service and Patient Records Database, behind their own authentication boundary and stricter access-control policy, separate from lower-sensitivity services like Notification. This follows the principle of least privilege: a compromise of the Notification Service, for example, does not expose prescription or patient record data, since each service authenticates independently (e.g. via service-to-service tokens/API gateway) and only the services that need patient health data can reach that database.

Performance Benefit

Because slot booking must feel instant to patients on a touch/web interface, the Appointment Booking Service can be deployed on more instances and closer to users (or cached/read-optimized for availability checks) without being slowed down by heavier, less time-sensitive operations such as generating and storing prescriptions or logging payment transactions, which can be processed asynchronously in the background.

Component Diagram


Telemedicine Slot Booking & Prescription Portal — Component Diagram
(Microservices architecture, subsystem view)
«subsystem» PatientPortal
internal structure
«subsystem» AppointmentService
internal structure
«subsystem» ClinicalRecords
internal structure
SearchSlots
:SlotSearch
Check Availability
:SlotManager
Update SlotStatus
BookAppointment
:BookingUI
Manage Consultation
:Consultations
Session
PatientLookup
Manage Patient
:PatientRecords
PatientLogin
:Authentication
PrescriptionAccess
:Prescriptions